

CHAPTER – VI

INHERITANCE:

- Inheritance means a **Subclass**(Child class) inherits all its properties and behaviour from its **Superclass**(Parent class).

ADVANTAGES:

- Code reusability
- Easy maintenance
- Extensibility
- Method overriding
- Organized code

TYPES:

- Single Inheritance : A ->B
- Multilevel Inheritance: A -> B -> C
- Hierarchical Inheritance : A ->B, A -> C
- Multiple Inheritance : A,B -> C
- Hybrid Inheritance : combination of above types.

SINGLE INHERITANCE:

- One class inherits from the another class
- A -> B.

EXAMPLE:

```
class Teacher{  
    void teach() {  
        System.out.println("The teacher can teach.");  
    }  
}
```

```

    }
}
class Sub extends Teacher{
    void maths() {
        System.out.println("The teacher can teach maths");
    }
}
public class SingInhe{
    public static void main(String [] args) {
        Sub s = new Sub();
        s.teach();
        s.maths();
    }
}

```

MULTILEVEL INHERITANCE:

- Class derived from another Subclass.
- A -> B -> C

EXAMPLE:

```

class Teacher1 {
    void teach() {
        System.out.println("The teacher can teach");
    }
}
class Subject extends Teacher {

```

```

        void maths() {
            System.out.println("The teacher can teach maths");
        }
    }
    class Incharge extends Subject{
        void staff() {
            System.out.println("The teacher is the incharge of class 7
");
        }
    }
    public class MulLevInhe{
        public static void main(String [] args) {
            Incharge i = new Incharge();
            i.teach();
            i.maths();
            i.staff();
        }
    }

```

HIEARARCHICAL INHERITANCE:

- Multiple class inherits from a single parent class.
- A -> B, A -> C

EXAMPLE:

```

class Student {
    void stu(){
        System.out.println("The Student Studies in the class 8");
    }
}

```

```
    }  
}  
class Mark1 extends Student{  
    void maths() {  
        System.out.println("He scored 90 marks in maths");  
    }  
}  
class Mark2 extends Student{  
    void science() {  
        System.out.println("He scored 89 marks in science");  
    }  
}  
public class hieararchicalInhe{  
    public static void main(String [] args) {  
        Mark1 m1 = new Mark1();  
        Mark2 m2 = new Mark2();  
        m1.stu();  
        m1.maths();  
        m2.stu();  
        m2.science();  
    }  
}
```

MULTIPLE INHERITANCE:

- One class inherits from two or more classes.
- This causes ambiguity (Diamond problem).
- A,B -> C

EXAMPLE:

```
interface Father{
    void fat();
}

interface Mother{
    void mot();
}

class Child implements Father, Mother{
    public void fat() {
        System.out.println("Ram is my father ");
    }
    public void mot() {
        System.out.println("Seetha is my mother");
    }
}

public class MultipleInhe{
    public static void main(String [] args) {
        Child c = new Child();
        c.fat();
        c.mot();
    }
}
```

HYBRID INHERITANCE:

- Combination of two or more types of inheritance.
- Java doesn't directly support hybrid inheritance through class but it allows via **interfaces**.

EXAMPLE:

```
interface Payment{
    void pay();
}

class Creditcard implements Payment{
    public void pay(){
        System.out.println("The account holder has a credit card");
    }
}

class Upi implements Payment{
    public void pay() {
        System.out.println("The account holder has both Credit
card and UPI");
    }
}

class OnlinePayment extends Upi{
    void display() {
        System.out.println("The account holder can use UPI for
online payment");
    }
}

public class HybridInhe {
```

```
public static void main(String [] args) {  
    OnlinePayment op = new OnlinePayment();  
    op.pay();  
    op.display();  
}  
}
```